

Write and Evaluate Algebraic Expressions

Lesson 6-5

Name: Type your name **Date:** Today's date **Class:** Class period



TODAY'S OBJECTIVES

Content Objective: I can write an algebraic expression for a real situation and evaluate it for a given value of the variable.

Language Objective: I can explain my expression using the words variable, term, coefficient, constant, substitute, and evaluate.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Write Algebraic Expressions

Variable

Algebraic Expression

Coefficient

Constant

Like terms

Substitute

Evaluate



Tap any word to see what it means and a picture.

1

Turn the words into numbers, operations and a

2

A letter that stands for an unknown number is a

3

A math phrase with variables, numbers, and operations but no equal sign is an

4

The number multiplied by a variable, like the 7 in $7n$, is the

5

A term that is just a number and never changes, like the 5 in $3x + 5$, is a

6 Terms with the same variable part are .

7 Replace the variable with a given number before you
 the expression.

8 To put numbers in for the letters and work out the value is to
 .

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

Write an expression for the total cost. What does each part of the expression represent?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Write Algebraic Expressions** — Translate words or a situation into numbers, variables, and operation symbols.

Escribir expresiones algebraicas — Traducir palabras o una situación a números, variables y símbolos de operaciones.

- ☐ **Variable** — A letter that stands for a number that is unknown or can change.

Variable — Una letra que representa un número desconocido o que puede cambiar.

- ☐ **Algebraic Expression** — A math phrase that has at least one letter.

Expresión algebraica — Una frase matemática que tiene al menos una letra.

- ☐ **Coefficient** — The number in front of a letter, like the 3 in $3x$.

Coeficiente — El número frente a una letra, como el 3 en $3x$.

- ☐ **Constant** — A fixed number in an expression or equation that does not change.

Constante — Un número fijo en una expresión o ecuación que no cambia.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

The total cost is ____ + ____h. The ____ represents the ____ and ____h represents the ____.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: Write $12t + 50$: the $12t$ grows with tickets, the 50 stays fixed.

Evidence: Students write $12t + 50$, identify t as the variable number of tickets, 12 as the coefficient (price per ticket), and 50 as the constant venue fee that does not change.

Reasoning: This evidence connects the definition of algebraic expression to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- The total cost is ____ + ____h. The ____ represents the ____ and ____h represents the ____.
- My answer is ____.

Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- My claim is ____.
- I know because ____.

Mi afirmación es ____.

Lo sé porque ____.

- So ____.

Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.
- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.
- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.